Yichao Cai

Ph.D. Candidate · Representation Learning & Multimodal AI

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Research Interests

Representation Learning · Multimodal/Multi-View Learning · Identifiability Analysis · Causal AI.

Research Statement

My research studies how modern learning objectives and supervision signals shape learned representations. I am particularly interested in when these objectives identify causal and generalizable latent structure, and when they instead discard, conflate, or leave it underdetermined. This perspective helps characterize the limits of foundation model objectives and distinguish capabilities that emerge through scaling from those requiring new objectives, supervision, or data interventions.

Methodologically, I use tools from identifiability theory, latent-variable modeling, population-objective analysis, and representation geometry. My broader goal is to develop a theory of representation learning that explains the capabilities and structural limits of modern learning systems.

Education

Ph.D. in Computer Science — Sep 2023 – Mar 2027

Australian Institute for Machine Learning, Adelaide University

Supervisor: Prof. Javen Qinfeng Shi · Expected thesis submission: Mar 2027

M.Sc. in Instrument Science and Technology — Sep 2016 – Jun 2019

Wuhan University of Technology

Supervisor: Prof. Xiao Zhou

B.Eng. in Measurement and Control Technology & Instrumentation — Sep 2012 – Jun 2016

Wuhan University of Technology

Publications

FIRST-AUTHOR AND CO-FIRST-AUTHOR PUBLICATIONS

[1] **Cai, Y.** and Shi, J. Q. "On the Identifiability of Masked Prediction: Mode Blindness and Mask Schedules." *arXiv:2608.01383*, 2026 (Preprint).

[2] **Cai, Y.**; Zhang, Z.; Liu, Y.; and Shi, J. Q. "The Geometric Mechanics of Contrastive Representation Learning: Alignment Potentials, Entropic Dispersion, and Cross-Modal Divergence." *International Conference on Machine Learning (ICML)*, 2026.

[3] **Cai, Y.***; Liu, Y.*; Gao, E.; Jiang, T.; Zhang, Z.; van den Hengel, A.; and Shi, J. Q. "On the Value of Cross-Modal Misalignment in Multimodal Representation Learning." *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. **Spotlight**.
*Equal contribution.

[4] **Cai, Y.**; Liu, Y.; Zhang, Z.; and Shi, J. Q. "CLAP: Isolating Content from Style Through Contrastive Learning with Augmented Prompts." *European Conference on Computer Vision (ECCV)*, 2024.

COLLABORATIVE PUBLICATIONS

[5] Liu, Y.; Gong, D.; **Cai, Y.**; Gao, E.; Zhang, Z.; Huang, B.; Gong, M.; van den Hengel, A.; and Shi, J. Q. "I Predict Therefore I Am: Is Next Token Prediction Enough to Learn Human-Interpretable Concepts from Data?" *International Conference on Learning Representations (ICLR)*, 2026.

Publications (continued)

[6] Jiang, W.; Liu, Y.; **Cai, Y.**; Gao, E.; Dong, J.; Abbasnejad, E.; Yao, L.; and Shi, J. Q. "What Makes a Representation Good for Single-Cell Perturbation Prediction?" *International Conference on Machine Learning (ICML)*, 2026.

[7] Chen, J.; Yin, Z.; **Cai, Y.**; Liu, Y.; Zhang, Z.; Gong, D.; and Shi, J. Q. "Boundary Embedding Shaping with Adaptive Contrastive Learning for Graph Structural Disentanglement." *International Conference on Machine Learning (ICML)*, 2026.

Ongoing Research

- **Predictive-state identifiability in next-token predictors:** investigating what predictive-state information finite-capacity next-token predictors must retain, and which aspects of latent structure remain underdetermined by the training objective.
- **Objective-induced equivalence classes of representation structure:** developing a framework for characterizing which representation structures are recoverable from data, supervision, and learning objectives.

Invited Talks

Understanding multimodal contrastive learning through two lenses: identifiability and population geometry — Jul 2026
Vision & Sensing Lab, Technion · Online invited talk
EML, Helmholtz Munich · Online invited talk

Teaching

ADELAIDE UNIVERSITY (formerly the University of Adelaide)

Teaching Assistant, Programming for Artificial Intelligence — Semester 2, 2026

Teaching Assistant, Neural Networks and Deep Learning — Semester 1, 2026

Guest Lecturer & Head Tutor, Statistical Machine Learning — Semester 2, 2025

Teaching Assistant, Using Machine Learning Tools — Trimester 2, 2025

Teaching Assistant, Concepts in AI and ML — Semester 1, 2025

Academic Service

Conference Reviewing AAAI 2027; NeurIPS 2026; ICML 2026 (Silver Reviewer Award); ICLR 2026.
Journal Reviewing Transactions on Machine Learning Research (TMLR).

Honors and Awards

The D. R. Stranks Travelling Fellowship — Jun 2026

NeurIPS 2025 Scholar Award, Neural Information Processing Foundation — Oct 2025

University of Adelaide Research Scholarship — Sep 2023

Outstanding Graduate Student, Wuhan University of Technology — Jun 2019

Additional Research Experience

Visiting Student Researcher — May 2018 – Sep 2018
California PATH, UC Berkeley

- Conducted research on visual perception algorithms for autonomous driving.